Explanation

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Experiment : Verification of Thevenin's Theorem

Objective

To verify **Thevenin's Theorem** by measuring full load current

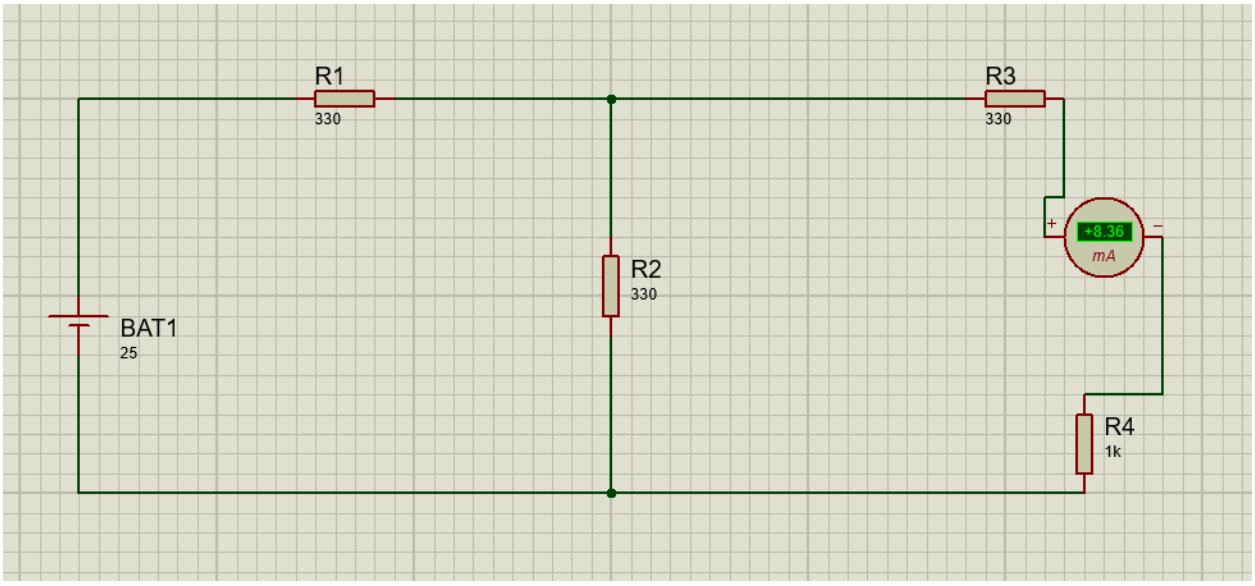
Theory

Thevenin's Theorem states that any linear, bilateral network consisting of multiple sources and resistors can be replaced by an equivalent circuit containing a single voltage source (Vth) in series with a single resistor This simplification allows for easy analysis of the circuit's behavior when the load resistance (Rth) is changed.

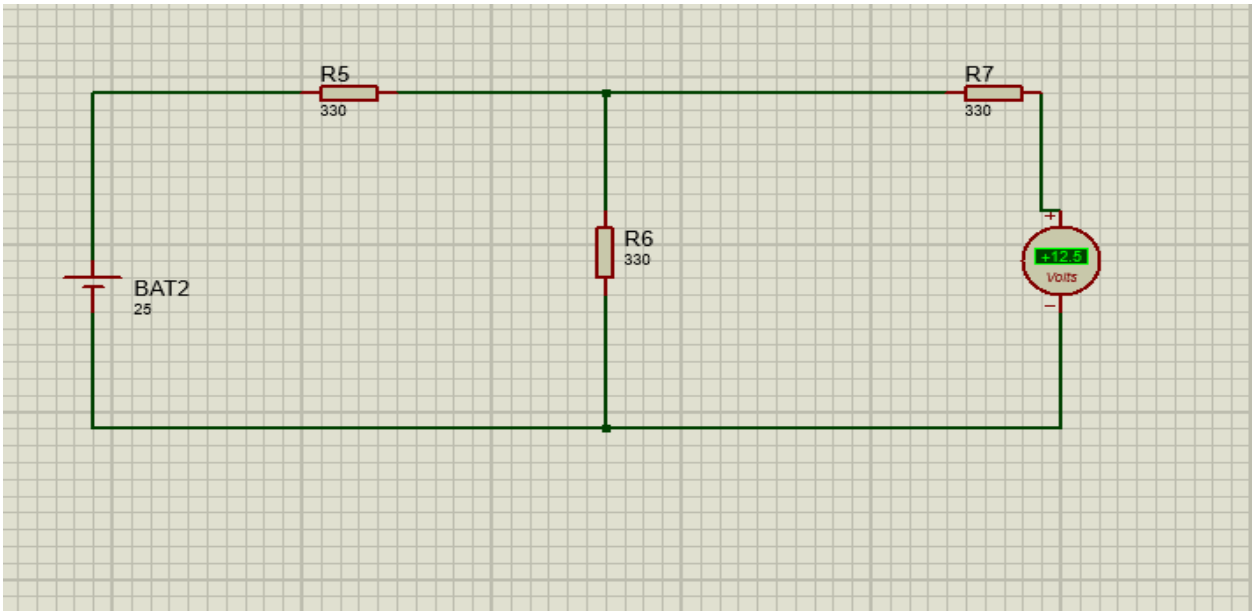
Procedure (Steps)

Step 1: Original Circuit Analysis

1. **Build the Circuit**: Place your components (Battery, R1,R2,R3,R4) and connect them
2. **Measure Load Current**: Place a DC Ammeter in series with the load resistor . Run the simulation and record the current value as IL for "Circuit - I".
3. **Note:** I have taken V as 25V

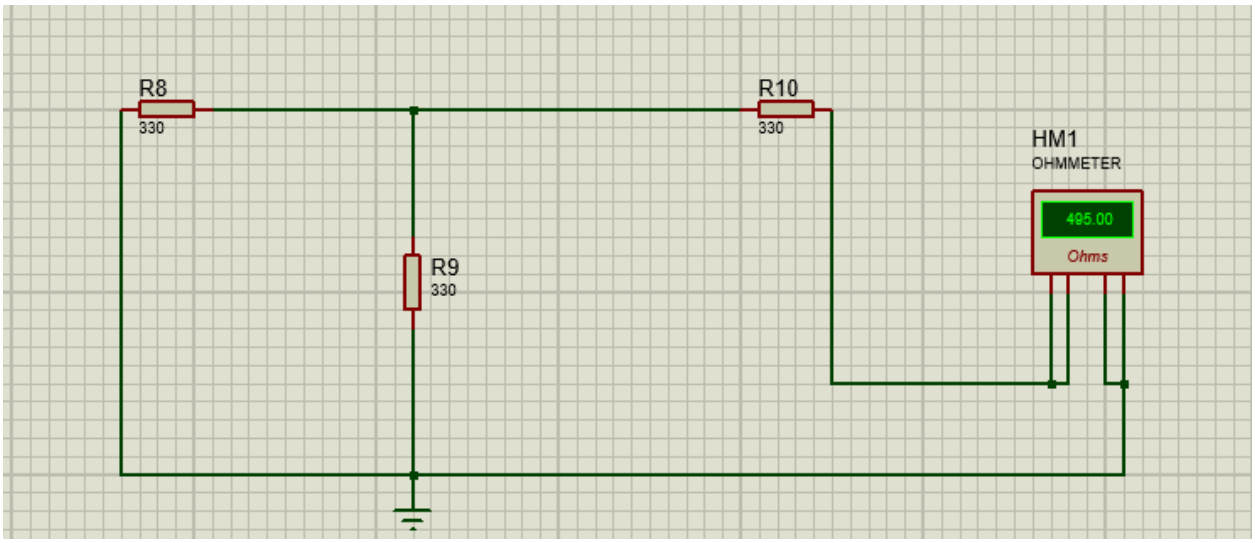


1. **Remove the Load:** Delete the load resistor R4 and the ammeter from the circuit.
2. **Measure Open-Circuit Voltage:** Place a **DC Voltmeter** across the open terminals where R4 used to be.
3. **Simulate:** Run the simulation. The reading on the voltmeter is your **Vth**



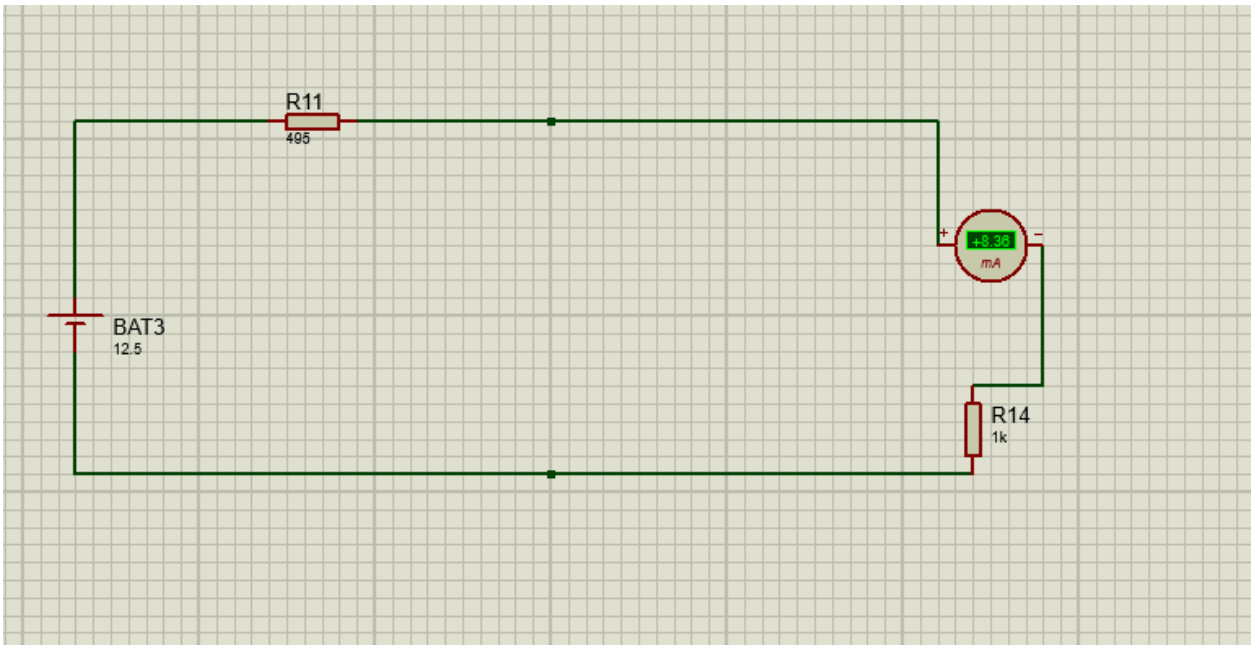
Step 3: Finding Rth (Thevenin Resistance)

1. **Deactivate Sources:** Replace the DC Battery with a short circuit (a plain wire).
2. **Measure Resistance:** Use an **Ohmmeter** component connected across the open load terminals.
3. **Simulate:** Run the simulation. The reading is your **Rth** .



1. **Build the Simplified Circuit:** Place a new DC Voltage source and set its value to your calculated Vth (**12.5V**).

2. **Add Rth:** Place a resistor in series with the source and set it to your calculated Rth 495 Ohm

3. **Reconnect Load:** Connect the original load resistor R4 1 Ohm and a DC Ammeter in series.

4. **Verify:** Run the simulation. The ammeter should show the same IL (approx **8.36 mA**) as the original circuit.